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DAFS: A distribution-aware hierarchical feature selection method for long-tailed classification

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ABSTRACT

Feature selection for long-tailed data has become a research hotspot due to high-dimensional features and imbalanced distributions in real-world data. Although some of them effectively balance the data, correctly classifying tail classes and distinguishing easy-confused classes in long-tailed data are still two significant challenges. To address these issues, we propose a distribution-aware hierarchical feature selection method for long-tailed classification (DAFS). First, we embed sample distribution-based punishment coefficients into loss and regularization terms to balance feature weights for head and tail classes, which enhances the accuracy of classifying tail classes. Then, we use multi-granularity knowledge and similarities among classes to design feature differentiation regularization terms for improving the distinguishability of easy-confused classes. Finally, extensive experimental results demonstrate that DAFS outperforms the other ten traditional and advanced feature selection methods on different datasets.

1. Introduction

Data often exhibit high-dimensional features and numerous classes owing to the expanding scope of applications [1]. High-dimensional features complicate computation and storage, posing challenges to traditional classification. Traditional feature selection methods aim to eliminate redundant and irrelevant features to solve high-dimensional problems and improve classification performance.

The increasing number of classes leads to long-tailed distributions and easy-confused classes, which pose challenges for traditional feature selection. A long-tailed distribution is an imbalanced scenario in which a few head classes contain many samples while most tail classes have significantly fewer samples. Traditional feature selection methods often prioritise features that benefit head class classification, neglecting the typically more critical tail classes. Easily confused classes refer to those with similar appearances, making them difficult to distinguish based on their shared features. For instance, wolves and dogs exhibit high similarity, leading to frequent misclassification.

Existing feature selection methods for addressing imbalance issues can be broadly categorised into data-level and algorithm-level approaches [2]. Data-level methods include oversampling [3], undersampling [4], and hybrid sampling techniques [5], which balance datasets by adjusting the number of samples in the tail or head classes. A common algorithm-level approach involves optimisation with regularised terms. For instance, Lu et al. [6] proposed a feature selection method that iteratively optimises to address high dimensionality and imbalanced data challenges. Du et al. [7] formulated a loss function within the large-margin framework, incorporating sample weights to mitigate class imbalance. While these methods enhance the importance of minority classes, they select a single shared feature subset for all classes, which may reduce interpretability in large-scale classification tasks.

Recently, hierarchical feature selection methods have gained significant attention for their ability to leverage multi-granularity knowledge to decompose large-scale tasks into subtasks and identify discriminative features for each subtask. Multi-granularity knowledge represents a hierarchical class structure, such as a tree or directed acyclic graph. Based on the tree structure, Zhao et al. [8] proposed an iterative optimization hierarchical feature selection method utilizing the sibling parent-child relation of classes. Based on the graph structure, Lin et al. [9] introduced a hierarchical feature selection based on label correlation in graph-structured data. Subsequently, methods for exploring the relationships among different classes are gradually proposed. For example, Liu et al. [10] proposed a hierarchical feature selection method that measures similarity between classes based on distance in the tree structure

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and class sample similarity. These methods demonstrate that feature selection guided by multi-granularity knowledge can identify superior and compact feature subsets for large-scale classification. However, existing hierarchical feature selection approaches overlook sample distribution, making them unsuitable for long-tailed data. Moreover, they lack effective strategies to address easily confused classes that are difficult to distinguish.

In this paper, we propose a distribution-aware hierarchical feature selection method using multi-granularity knowledge for long-tailed data (DAFS). Specifically, we first embed penalty coefficients based on sample distribution into the loss and regularization terms to adjust the feature weights of the head and tail classes. This distribution-aware weight adjustment increases the weight of features beneficial for correctly classifying tail classes, minimizing the misclassification of tail classes. Then, we introduce a feature differentiation regularization term that considers multi-granularity knowledge and uses the similarity between classes, which increases the distinguishability of easily confused classes. Our method identifies optimal feature subsets for large-scale imbalanced data, which can be applied in long-tailed classification to improve the classification accuracy of tail classes and alleviate the problem of easily confused classes being misclassified.

The main contributions of this paper are as follows: (1) From a sample perspective, we incorporate distribution-based penalty coefficients to adjust feature weights. This strategy reduces the risk of misclassifying tail classes while balancing the importance of features between head and tail classes, thereby improving tail class classification accuracy. (2) From a class perspective, we utilise multi-granularity structural information to decompose the complex classification task and enhance feature differentiation between easily confused classes by leveraging class similarities. (3) Extensive experiments on various long-tailed datasets demonstrate the competitive performance of DAFS compared with ten traditional and advanced methods.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 details the proposed framework and its optimisation process. Section 4 outlines the experimental settings, while Section 5 presents the experimental results and analysis. Finally, Section 6 summarises the study and discusses future research directions.

2. Related work

In this section, we briefly review related work, covering traditional feature selection, feature selection for imbalanced data, and feature selection driven by structural knowledge.

2.1. Traditional feature selection

Feature selection is a data preprocessing technique to extract a representative and discriminative subset of features from the original dataset [11]. It is particularly important for high-dimensional data, as it helps reduce computational complexity and mitigate overfitting. One prominent approach is embedded feature selection, which aims to identify an optimal feature subset by optimising an objective function. This approach typically includes two key strategies: redundancy minimisation-based methods and manifold learning-based methods. The former enhances feature differentiation by reducing redundancy. For instance, Li et al. [12] assigned weights to selected features by incorporating their similarity, ensuring low redundancy among features. Jia et al. [13] utilised the $l_{2,1}$-norm and the inner product of feature weight vectors to eliminate redundant features.

Manifold learning-based methods leverage locality to establish relationships between objects. Sun et al. [14] explored local label correlations through manifold learning frameworks to enhance label discrimination. Zhang et al. [15] utilised the low-dimensional manifold structure of pseudo labels to constrain feature weights. Furthermore, Shang et al. [16] applied dual manifold regularisation to preserve local structural information across different spaces, effectively capturing global and local features.

However, these traditional feature selection methods focus solely on relationships within the feature space, overlooking the imbalanced class distribution present in real-world data.

2.2. Feature selection for imbalanced data

Feature selection methods for imbalanced data have gained significant attention as a means of addressing the challenges posed by skewed class distributions. These methods can be broadly categorised into data-level and algorithm-level approaches. Data-level methods achieve balance by adjusting the number of samples, primarily through oversampling, undersampling, and hybrid sampling techniques. Oversampling increases the number of samples in tail classes to match the distribution of head classes. For instance, Tsai et al. [3] addressed data imbalance by generating minority class samples and applying attribute reduction. In contrast, undersampling balances the dataset by reducing the number of samples in the head class. Abujazoh et al. [4] addressed class imbalance by randomly removing samples from the majority classes. Hybrid sampling combines undersampling and oversampling techniques to achieve a more balanced dataset. Sun et al. [5] proposed a feature selection method that applies hybrid sampling to imbalanced data by considering the distance between clusters. Although these methods effectively balance the dataset, they alter the original data distribution, which can result in information loss and an increased risk of overfitting.

Algorithm-level methods address imbalanced datasets without modifying the number of samples in different classes. Han et al. [17] employed a multi-objective fireworks algorithm to select multiple equivalent feature subsets, effectively tackling classification challenges in imbalanced datasets. Kamalov et al. [18] introduced a decision tree-based approach to mitigate the tendency of traditional feature selection methods to favour majority classes in imbalanced data. Some methods tackle class imbalance by optimising regularisation terms. For example, Lu et al. [6] proposed a feature selection method that integrates manifold learning with label imbalance processing to address challenges in imbalanced data. Du et al. [7] designed a loss function incorporating sample weights to mitigate class imbalance. These methods directly address imbalance issues by enhancing the importance of minority classes without altering the original data distribution.

However, the methods above assume that all classes are independent without considering their relationships. The methods overlook the potential multi-granularity knowledge inherent in class structures, making them unsuitable for large-scale hierarchical classification tasks. In contrast, the proposed method decomposes a large-scale task into subtasks by partitioning data with multi-granularity knowledge, reducing the complexity of classification.

2.3. Feature selection driven by multi-granularity knowledge

Multi-granularity structural knowledge has been widely applied in feature selection because it can capture complex hierarchical class relationships. Feature selection guided by multi-granularity structural knowledge can be categorised into global and local hierarchical feature selection. The former focuses on inter-class relationships to improve overall classification performance. For example, Huang et al. [19] proposed a feature selection method that integrates semantic label knowledge with spatial structural relationships. Huang et al. [20] introduced a hierarchical incremental feature selection method that accounts for the dynamic nature of real-world data. While these approaches utilise multi-granularity structural knowledge to explore global class relationships, they fail to capture internal relationships at the local granularity.

To address this limitation, local hierarchical feature selection decomposes the original task into subtasks of varying granularity and selects relevant features for each. According to different structural representations, local hierarchical feature selection can be categorised into

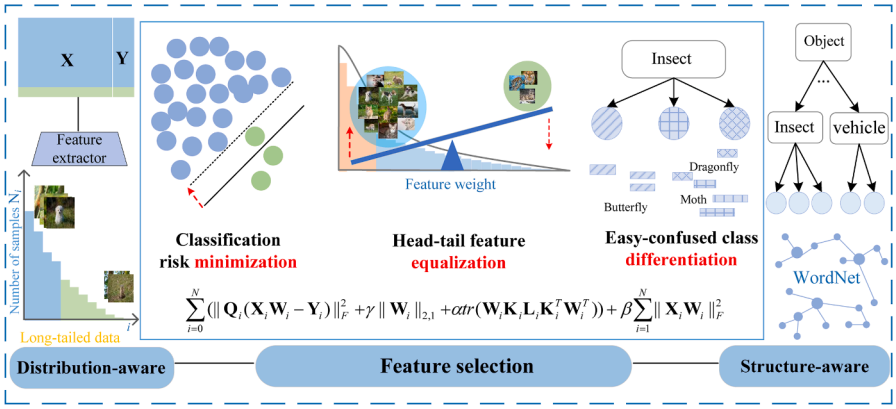


Fig. 1. The DAFS framework.

graph-based and tree-based approaches. From the perspective of graph structures, Lin et al. [9] proposed a feature selection algorithm that identifies common and class-specific features by leveraging label correlations in complex hierarchical graph-structured data. From the perspective of tree structures, Zhao et al. [8] proposed a hierarchical feature selection method based on recursive regularisation. This method involves the iterative selection of feature subsets for each subtask within the tree structure. Subsequently, Liu et al. [10] introduced a feature selection method that leverages sibling classes to provide additional supervisory information for each subtask. Zhao et al. [21] proposed a hierarchical feature selection method with adaptive sample weighting to reduce the impact of incorrect samples in each subtask.

However, the methods above overlook sample distribution-based class frequency information and fail to improve tail class accuracy. The proposed method addresses this limitation by adjusting feature weights based on class frequency to enhance tail class classification accuracy while leveraging multi-granularity knowledge to mitigate the challenge of distinguishing easily confused classes.

3. Proposed method

In this section, we elaborate on the distribution-aware feature selection framework for long-tailed data driven by multi-granularity knowledge (DAFS).

3.1. Framework of DAFS

DAFS divides the task into subtasks of varying granularity to address simpler and more balanced problems. Additionally, it coordinates the feature weights of the head and tail classes through distribution information and differentiates the features of easily confused classes by multi-granularity knowledge. The framework of DAFS is depicted in Fig. 1, and it can be divided into the following three main parts:

(1) Classification risk minimization: DAFS embeds distribution-aware mitigation factors into the loss term to enhance the attention for tail classes and reduce the misclassification risk of tail classes.

(2) Head-tail feature equalization: A moderator factor is added to the manifold learning regularization term to equalize the features of the head and tail classes.

(3) easily confused class differentiation: A discriminative regularization term based on class similarity increases the distinguishability of easily confused classes.

3.2. Classification risk minimization

DAFS aims to select related features for the tail classes that are overwhelmed and ignored by the head classes. Extreme class imbalance in large-scale classification tasks results in an over-preference for head classes at the expense of tail classes [22]. We use a multi-granularity abstract knowledge structure to decompose large-scale global tasks (classifying all classes) into local subtasks, which can reduce the imbalance and complexity between classes in the original task. Specifically, we divide samples belonging to the same multi-granularity node into the same subtask. Specifically, we divide samples belonging to the same multi-granularity node into the same subtask. Assuming the feature matrix of the ith task is expressed as $\mathbf{X}_i = [\mathbf{x}_i^1; \cdots; \mathbf{x}_i^j; \cdots; \mathbf{x}_i^{n_i}] \in \mathbb{R}^{n_i \times m}$, where $i \in \{0, 1, 2 \cdots, N\}$, n_i is number of samples in the ith task, and m is the feature dimension. The label matrix is represented as $\mathbf{Y}_i = [\mathbf{y}_i^1; \cdots; \mathbf{y}_i^j; \cdots; \mathbf{y}_i^{n_i}] \in \{0, 1\}^{n_i \times c_i}$, where c_i indicates the number of classes in the ith task. The feature weight matrix is present as $\mathbf{W}_i = [\mathbf{w}_i^1; \cdots; \mathbf{w}_i^j; \cdots; \mathbf{w}_i^m] \in \mathbb{R}^{m \times c_i}$, where the value of $\mathbf{w}_i^j$ measure the importance of the jth feature in the ith task. The importance degree of each feature in the ith task is determined by sorting the values of $||\mathbf{W}_i||_2$, then selecting the top few of them to achieve dimensionality reduction.

The basic paradigms of feature selection are embedded in each subtask. First, we embed a distribution-aware mitigation factor for the balanced loss term. Samples of tail classes tend to be more critical and valuable than head classes in real-world applications, and their classification error may lead to more serious consequences. This implies that the risk of misclassifying tail classes as head classes is far greater than that of the reverse. Therefore, we add mitigation weights to the loss function to increase the weight of tail classes in the loss function and balance the distribution of the loss function. The mitigation weight matrix of the ith task, which is a diagonal matrix with diagonal elements defined as follows:

$$Q_i^{jj} = \frac{\sum_{k=1}^{c_i} S_i^k - S_i^j}{S_i^j} + r, \tag{1}$$

where S_i^j is the sample number for the jth class of the ith task, $\frac{\sum_{k=1}^{c_i} S_i^k - S_i^j}{S_i^j}$ is normalized, and r is a parameter used to fine tune the value of Q_i^{jj}. The least squares loss is often used to minimize the squared error between predicted labels and true labels, enabling the model to learn the parameters closest to the true labels. We embed the mitigation factor in the loss function to minimize classification risk as in Fig. 2(b).

Then, we utilize the $l_{2,1}$-norm to sparsify the features due to the plentiful redundant and irrelevant features[23]. Finally, the equation for minimizing the classification risk based on balanced loss is defined as follows:

$$\min_{\mathbf{W}_0,\cdots,\mathbf{W}_N} \sum_{i=0}^{N} ||\mathbf{Q}_i(\mathbf{X}_i\mathbf{W}_i - \mathbf{Y}_i)||_F^2 + \gamma\|\mathbf{W}_i\|_{2,1}, \tag{2}$$

where γ is the parameter to control the impact of sparsity, and the symbol $||\cdot||_F$ represents the Frobenius norm. The weighted least squares loss changes the direction of feature weight optimization, focusing more on the prediction error of tail classes. This improves the classification

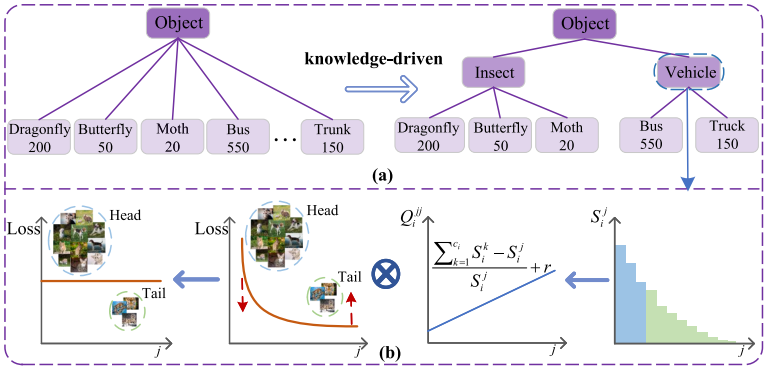


Fig. 2. (a) An example of the task partitioning; (b) The influence of mitigation factors on loss.

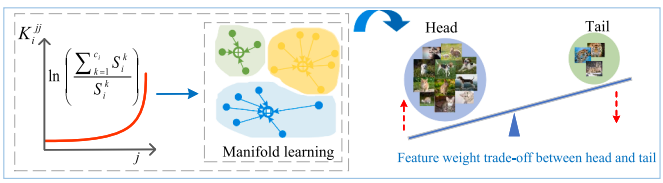


Fig. 3. Equalization of head and tail feature weights.

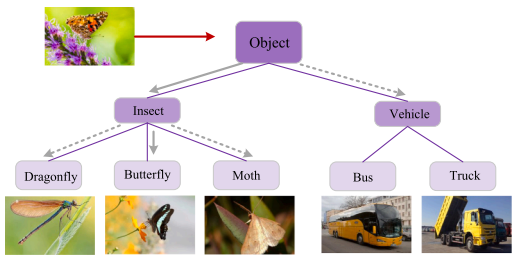


Fig. 4. An example of the classification process for an easy-confused class.

accuracy of tail classes while reducing the risk of misclassification. Example 1 explains the process of task decomposition driven by multi-granularity knowledge.

Example 1. As shown in Fig. 2(a), the coarse-grained task is divided into subtasks from top to bottom inspired by human "concrete-abstract" thinking. The degree of task imbalance and scale relatively decreases as tasks are divided. The sample "Butterfly" is initially classified into the finer-grained class "Insect" and then subdivided into "Moth", "Butterfly" or "Dragonfly" in a relatively balanced task.

3.3. Head-tail feature equalization

In long-tailed data, discriminative features for tail classes are often overshadowed by those of head classes. Generally, most features beneficial for the head class are easily regarded as important and given greater weight, while only a few features that benefit tail classes are selected. To address the issue of ignoring the important features for tail classes, we design an adjustment matrix $\mathbf{K}_i$ to rebalance the feature weight between head and tail. The diagonal element K_i^{jj} of $\mathbf{K}_i$ is calculated as follows:

$$K_i^{jj} = \ln\left(\frac{\sum_{k=1}^{c_i} S_i^k}{S_i^j}\right). \tag{3}$$

For the ith task, we embed matrix $\mathbf{K}_i$ into manifold learning to balance the weights of head and tail features while maintaining the similarity of the data in low-dimensional space, as shown in Fig. 3.

The head-tail feature balancing regularizer is as follows:

$$tr(\mathbf{W}_i\mathbf{K}_i\mathbf{L}_i\mathbf{K}_i^T\mathbf{W}_i^T), \tag{4}$$

where $\mathbf{L}_i = \mathbf{M}_i - \mathbf{P}_i$, $\mathbf{P}_i$ is the class similarity matrix by calculating the cosine similarity between the instance means of the classes, and $\mathbf{M}_i$ is a diagonal matrix with its diagonal element is $M_i^{jj} = \sum_{k=1}^{c_i} P_i^{jk}$. The symbol $tr(\cdot)$ represents the trace of a matrix. In Eq. (4), K_i^{jj} enhances the optimization of tail class features by reverse-weighting the sample size. This forces the optimization process to focus more on the tail class features, thereby separating them from head class features and preventing the tail class features from being overwhelmed, preserving discriminative features for the tail class.

After adding the regularization term for balancing the weights of head and tail features, the new objective function is as follows:

$$\min_{\mathbf{W}_0,\cdots,\mathbf{W}_N} \sum_{i=0}^{N} ||\mathbf{Q}_i(\mathbf{X}_i\mathbf{W}_i - \mathbf{Y}_i)||_F^2 + \alpha tr(\mathbf{W}_i\mathbf{K}_i\mathbf{L}_i\mathbf{K}_i^T\mathbf{W}_i^T) + \gamma\|\mathbf{W}_i\|_{2,1}, \tag{5}$$

where α is the trade-off parameter for feature weight between the head and tail classes.

3.4. Easy-confused class differentiation

The significant increase in classes has led to the growth of easy-confused classes. Classes under the same parent class node in the hierarchy usually exhibit greater similarities, which are susceptible to confusion and misclassifying. Distinguishing the easy-confused classes becomes increasingly challenging, as explained in Example 2 below.

Example 2. As shown in Fig. 4, there is a significant difference between "Insect" and "Vehicle". They are easily distinguished by one feature, such as "Wheels". However, classification is more intricate for the easy-confused class in fine-grained tasks. For example, distinguishing between "Butterfly" and "Moths" with similar appearances requires classification based on more detailed features, such as "Wing color". It is essential to differentiate their features to classify the easy-confused classes accurately.

There are similar features among easy-confused classes, leading to classification relying on redundant features and making it difficult to distinguish. To distinguish easy-confused classes, we design the feature differentiation regularization term:

$$tr(\mathbf{X}_i^T\mathbf{X}_i\mathbf{W}_i\mathbf{W}_i^T). \tag{6}$$

The feature differentiation regularization term amplifies the weight of discriminative features by penalizing redundant similar features. When highly correlated features exhibit strong similarity, Eq. (6) enforces their corresponding weights $\mathbf{w}_j$ and $\mathbf{w}_k$ tending towards orthogonality, thereby resisting the joint effect of similar features.

After adding feature differentiation regularization terms for easy-confused classes, the final target equation is as follows:

$$J(\mathbf{W}_0,\cdots,\mathbf{W}_N)=\min_{\mathbf{W}_0,\cdots,\mathbf{W}_N}\sum_{i=0}^{N}\Bigg(\underbrace{||\mathbf{Q}_i(\mathbf{X}_i\mathbf{W}_i-\mathbf{Y}_i)||_F^2}_{\text{Classification risk minimization}}+\underbrace{\alpha tr(\mathbf{W}_i\mathbf{K}_i\mathbf{L}_i\mathbf{K}_i^T\mathbf{W}_i^T)}_{\text{Head-tail feature regulation}}+\underbrace{\gamma\|\mathbf{W}_i\|_{2,1}}_{\text{Feature sparsification}}\Bigg)+\underbrace{\beta\sum_{i=1}^{N}tr(\mathbf{X}_i^T\mathbf{X}_i\mathbf{W}_i\mathbf{W}_i^T)}_{\text{Easy-confused class differentiation}}, \tag{7}$$

where β is the parameter of the feature differentiation term.

Overall, we embed the class structure into regularization terms, and the mitigation factor based on sample distribution is embedded into the loss function and regularization term. These strategies can minimize classification risk and improve the tail class accuracy. In addition, the feature differentiation term is introduced to increase the distinguishability of easy-confused classes.

3.5. Optimization

This section describes the optimization process for the final optimization objective Eq. (7). Due to the non-smoothness of the $l_{2,1}$-norm paradigm, we according to [24] and obtain as folows:

$$\frac{\partial(||\mathbf{W}_i||_{2,1})}{\partial\mathbf{W}_i}=\frac{\partial(tr(\mathbf{W}_i{}^{\mathrm{T}}\mathbf{D}_i\mathbf{W}_i))}{\partial\mathbf{W}_i}=2\mathbf{D}_i\mathbf{W}_i, \tag{8}$$

where $\mathbf{D}_i\in\mathbb{R}^{m\times m}$ is a diagonal matrix in which the diagonal element is defined as:

$$d_i^{jj}=\frac{1}{2||\mathbf{w}_i^j||_2}. \tag{9}$$

For the root node in the multi-granularity tree structure, by setting the derivative of Eq. (7) concerning $\mathbf{W}_0$ to zero, we have

$$\frac{\partial J(\mathbf{W}_0)}{\partial\mathbf{W}_0}=\mathbf{X}_0^T\mathbf{Q}_0^T\mathbf{Q}_0\mathbf{X}_0\mathbf{W}_0-\mathbf{X}_0^T\mathbf{Q}_0^T\mathbf{Q}_0\mathbf{Y}_0+\alpha\mathbf{W}_0\mathbf{K}_0\mathbf{L}_0\mathbf{K}_0^T+\gamma\mathbf{D}_0\mathbf{W}_0. \tag{10}$$

Therefore, we have the expression for $\mathbf{W}_0$ as

$$(\mathbf{X}_0^T\mathbf{Q}_0^T\mathbf{Q}_0\mathbf{X}_0+\gamma\mathbf{D}_0)\mathbf{W}_0+\alpha\mathbf{W}_0\mathbf{K}_0\mathbf{L}_0\mathbf{K}_0^T=\mathbf{X}_0^T\mathbf{Q}_0^T\mathbf{Q}_0\mathbf{Y}_0. \tag{11}$$

Eq. (11) is a Sylvester equation, whose form is consistent with $\mathbf{AX}+\mathbf{XB}=\mathbf{C}$. We use the Sylvester function to solve this equation according to [6].

For the internal nodes in the multi-granularity tree structure, by setting the derivative of Eq. (7) with respect to $\mathbf{W}_i$ to zero, we have

$$(\mathbf{X}_i^T\mathbf{Q}_i^T\mathbf{Q}_i\mathbf{X}_i+\beta\mathbf{X}_i^T\mathbf{X}_i+\gamma\mathbf{D}_i)\mathbf{W}_i+\alpha\mathbf{W}_i\mathbf{K}_i\mathbf{L}_i\mathbf{K}_i^T=\mathbf{X}_i^T\mathbf{Q}_i^T\mathbf{Q}_i\mathbf{Y}_i. \tag{12}$$

Similarly, we employ the Sylvester function to solve Eq. (12). During iteration, $\mathbf{W}_i$ continuously updates the optimization until convergence. The specific process steps of DAFS are shown in Algorithm 1.

The computational complexity of the DAFS algorithm mainly depends on the update of the feature weight matrix in line 9. This involves solving the Sylvester equation in Eq. (12). According to [25], we use the Bartels-Stewart algorithm to perform this step and obtain a time complexity of $O(m^3+k^3)$, where m and c_i are the number of features and classes, respectively. Generally, the class dimension of data is small and can be ignored. Therefore, the time complexity of DAFS is $O(m^3)$.

4. Experimental settings

In this section, we present the details of the experimental setup, including the following four aspects: (1) Dataset description, (2) Comparison methods, (3) Evaluation metrics, and (4) Implementation details.

Algorithm 1 The algorithm of DAFS.

Input: (1) Feature matrix $\mathbf{X}_i$ and label matrix $\mathbf{Y}_i$; (2) Regularization parameters: α, β, and γ; (3) T is the maximum number of iterations;
Output: Weight matrices $\mathbf{W}=[\mathbf{W}_0,\mathbf{W}_1,\cdots,\mathbf{W}_N]$. Sorting features according to the descending order of $||\mathbf{W}_i||_2$ and selecting top k features as feature sets of the ith subtasks.

1: Set iteration number $t=1$;
2: Initialize $\mathbf{W}=[\mathbf{W}_0,\mathbf{W}_1,\cdots,\mathbf{W}_N]$ randomly;
3: **for** $i=0:N$ **do**
4: Compute the diagonal matrix $\mathbf{D}_i$ according to Eq. (9);
5: **end for**
6: **while** $t\le T$ **do**
7: Update $\mathbf{W}_0$ by solving Eq. (11);
8: **for** $i=1:N$ **do**
9: Update $\mathbf{W}_i$ by solving Eq. (12);
10: **end for**
11: Update $\mathbf{W}^{(t+1)}=[\mathbf{W}_0^{(t+1)},\mathbf{W}_1^{(t+1)},\cdots,\mathbf{W}_N^{(t+1)}]$;
12: $t=t+1$;
13: **end while**
14: **return** $\mathbf{W}$;

Table 1
Description of the datasets.

Dataset	F194	VOC	SUN	CLEF	Cifar100	ILSVRC
Imbalance ratio	42.1	57	29.9	383.6	1.3	1.4
Feature	473	4096	4096	4096	4096	4096
Training	7105	3437	45,109	6405	50,000	12,346
Test	1420	3539	22,556	3202	10,000	11,845
Node	202	30	343	88	121	65
Leaf (Class)	194	20	324	80	100	57
Type	Protein	Image	Image	Image	Image	Image

4.1. Dataset description

We divided all datasets in the experiment into long-tailed datasets and balanced datasets. The long-tailed datasets comprise F194 [26], VOC [27], SUN [28], and CLEF [29]. F194 comprises 8525 samples and 473 features, with 194 real classes. VOC is a standard image recognition and object detection dataset containing 20 object classes, which spill 3437 training and 3539 test samples, each represented by 4,096-dimensional features. SUN encompasses 324 fine-grained scene classes with 45,109 training and 22,556 test samples, with 4096 features. CLEF consists of 8368 training samples, 939 test samples, and 80 real classes. The balanced datasets comprise Cifar100 [30] and ILSVRC [31]. Cifar100 is a widely used image classification benchmark dataset consisting of 50,000 training samples, 10,000 test samples, 100 classes, and 4096 features. ILSVRC is a large-scale benchmark with 57 object classes comprising 12,346 training and 11,845 test samples. The details about the datasets are shown in Table 1.

The imbalance ratio of the dataset is represented by ρ, which is calculated as $\rho=\frac{n_{\max}}{n_{\min}}$, where $n_{\max}$ and $n_{\min}$ represent the number of samples in the most frequent and least frequent classes, respectively. We construct the long-tailed variants of the Cifar100 dataset according to the reduced samples for each class in the training set based on the imbalance ratio $\rho=\{50,100,200\}$. The three synthetic long-tailed Cifar100-LT datasets are delineated in Table 2. Fig. 5 demonstrates the distribution of class-wise samples in the imbalanced long-tailed datasets.

4.2. Comparison methods

We leverage ten comparison algorithms to verify the effectiveness of DAFS, including modified traditional and advanced methods. The following are descriptions of the comparison algorithms:

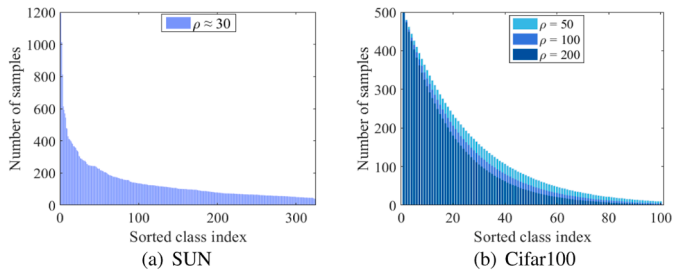


Fig. 5. Histograms of the number of class samples on different datasets.

Table 2
Description of Cifar100-LT with different imbalance ratios.

Dataset	Cifar100-LT		
Imbalance ratio	$\rho = 50$	$\rho = 100$	$\rho = 200$
Feature	4096	4096	4096
Training	12,608	10,847	9502
Test	10,000	10,000	10,000
Node	121	121	121
Leaf (Class)	100	100	100

(1) **HMRMR** [32] incorporates the principles of minimizing redundancy and maximizing relevance into the hierarchical feature selection. (2) **Hfisher** [33] selects the most discriminative features by maximizing the separation between classes and minimizing the cohesion of classes, which embeds the idea of Fisher into the hierarchical structure. (3) **HFSNM** [23] combined with the $l_{2,1}$-norm-based loss function and regularization term to select for features with joint sparsity. (4) **HFSRR** [8] selects the optimal subset of features for each granularity iteratively using the inter-granularity relationship. (5) **HFSLDL** [10] mitigates the class imbalance problem by hierarchical label distributions using the relationship of classes in the tree structure. (6) **HFSCN** [34] is a hierarchical feature selection method using the capped l_2-norm based loss to alleviate the issue of error propagation across layers. (7) **HFSDK** [35] is a hierarchical feature selection method that is robust to data outliers. (8) **HFSLE** [36] is a hierarchical feature selection method with label enhancement to address semantic gaps. (9) **HFSCF** [37] considers consistency within granularity and differences between sibling granularities to select representative features. (10) **PHFS** [21] is an adaptive sample weighting-based progressive hierarchical feature selection, which dynamically adjusts the sample weights to focus on high-quality data.

4.3. Evaluation metrics

We use four evaluation metrics to measure the classification performance, including ACC, TIE, F_H, and F_{LCA}. (1) ACC is a flat metric widely used to evaluate the overall classification accuracy. (2) TIE [38] is the number of edges in the shortest path between the real and predicted classes. (3) F_H [39] is an ensemble-based evaluation metric that considers all common ancestors between the true and predicted classes of a sample. (4) F_{LCA} [40] considers the nearest common ancestor between the true and predicted class of a sample, thus avoiding the influence of too many ancestor nodes.

4.4. Implementation details

Classifier: For all the experiments, the performance of hierarchical classification algorithms is evaluated by KNN (K-Nearest Neighborhood), SVM (Support Vector Machine), and RF (Random Forest) classifiers.

Parameter setting: We set parameters $\alpha = 0.1$ and $\beta = 100$ for the protein datasets and parameters $\alpha = 0.1$ and $\beta = 0.1$ for the image datasets. The parameter r is seted to 1 in the experiment.

Local class splitting: The classes in the training set are divided into three class subsets based on the number of samples: head classes (more than 100 samples), middle classes (20–100 samples), and tail classes (1–20 samples).

The experiments uniformly select 10% of the protein dataset features and 20% of the image dataset features as feature subsets to compare performance. Experimental results are averaged under a 10-fold cross-validation strategy.

All experiment results are implemented on a computer with an Intel(R) Xeon(R) Gold 6126, 2.60 GHz CPU, and the 64-bit Windows 10 operating system. The code can be downloaded from the following link: https://github.com/fhqxa/DAFS.

5. Experimental results and analysis

In this section, we design experiments to verify the effectiveness of the DAFS method in six aspects: (1) effectiveness comparison with other methods; (2) comparison of local classification accuracy; (3) visualization; (4) ablation analysis; (5) effectiveness analysis of feature differentiation regularizer; and (6) convergence analysis.

5.1. Effectiveness comparison with other methods

In this section, we compare seven state-of-the-art hierarchical feature selection methods and three traditional methods modified with hierarchy on different datasets, respectively. The experiment uses SVM, KNN, and RF classifiers to test performance and evaluates results based on ACC, F_H, F_{LCA}, and TIE metrics.

The TIE comparison results for DAFS and traditional methods are depicted in Table 3. From this table, it is evident that DAFS outperforms other traditional methods across three classifiers. The TIE value is superior to other methods except for the small dataset F194 on the RF classifier. This indicates that our method outperforms traditional methods in hierarchical classification.

The ACC, F_H, and F_{LCA} results of the experiments compared with seven advanced methods are listed in Tables 4–6. According to these results, the following conclusions can be obtained:

(1) The performance of DAFS is generally better than that of most methods on three classifiers. Compared with the state-of-the-art method PHFS, DAFS ranks second only on the VOC dataset but ranks first on all other datasets. Overall, DAFS achieves a competitive and stable performance compared with the other methods.

(2) In the small dataset F194, the F_H value of DAFS on the SVM classifier is slightly lower than HFSRR but still ranks second. The complexity of the hierarchical subtasks in F194 could result in classification errors.

Table 3
TIE comparison between DAFS and three methods on different classifiers (↓).

Classifier	Method	Dataset						Avg.
		F194	VOC	SUN	CLEF	Cifar100	ILSVRC	
SVM	HMRMR	0.1785	0.1155	0.1322	0.1825	0.1273	0.0335	0.1282
	Hfisher	0.1894	0.1159	0.1341	0.2011	0.1285	0.0336	0.1338
	HFSNM	0.2096	0.1191	0.1351	0.2007	0.1329	0.0350	0.1388
	DAFS	**0.1706**	**0.1149**	**0.1293**	**0.1781**	**0.1239**	**0.0316**	**0.1247**
KNN	HMRMR	0.2017	0.1044	0.1552	0.1782	0.1324	0.0547	0.1378
	Hfisher	0.1925	0.1040	0.1529	0.1951	0.1320	0.0549	0.1386
	HFSNM	0.2048	0.1110	0.1732	0.1935	0.1415	0.0602	0.1474
	DAFS	**0.1851**	**0.1025**	**0.1487**	**0.1756**	**0.1294**	**0.0512**	**0.1321**
RF	HMRMR	0.1558	0.1029	0.1314	0.1772	0.1741	0.0332	0.1291
	Hfisher	0.1437	0.1030	0.1319	0.1882	0.1333	0.0331	0.1222
	HFSNM	**0.1349**	0.1113	0.1363	0.1906	0.1378	0.0375	0.1248
	DAFS	0.1387	**0.1022**	**0.1278**	**0.1741**	**0.1308**	**0.0330**	**0.1178**

Table 4
ACC comparison among seven methods on different classifiers (↑).

Classifier	Dataset	Method							
		HFSRR	HFSLDL	HFSCN	HFSDK	HFSLE	HFSCF	PHFS	DAFS
SVM	F194	0.3430	0.3409	0.3401	0.3275	0.3416	0.3423	0.3232	**0.3444**
	VOC	0.5841	0.5849	0.5818	0.5818	0.5824	0.5815	0.5869	**0.5897**
	SUN	0.6280	**0.6307**	0.6250	0.6216	0.6257	0.6282	0.6132	0.6234
	CLEF	0.6028	0.6060	0.5986	0.5847	0.6103	0.6177	0.5368	**0.6219**
	Cifar100	0.6186	0.6199	0.6147	0.6205	0.6212	0.6182	0.6144	**0.6271**
	ILSVRC	0.8498	0.8506	0.8241	0.8232	0.8563	0.8480	0.8520	**0.8559**
	Avg.	0.6165	0.6175	0.6065	0.6065	0.6177	0.6179	0.5878	**0.6200**
KNN	F194	0.2972	0.2901	0.3035	0.2909	0.2803	0.2944	0.2655	**0.3049**
	VOC	0.6123	0.6183	0.6118	0.6146	0.6191	0.6135	0.6310	**0.6293**
	SUN	0.5376	0.5369	0.5294	0.5274	0.5432	0.5381	0.5311	**0.5411**
	CLEF	0.6304	**0.6315**	0.6092	0.6209	0.6144	0.6251	0.5697	0.6091
	Cifar100	0.5935	0.5908	0.5850	0.5887	0.5952	0.5912	0.6021	**0.6085**
	ILSVRC	0.7390	0.7412	0.6603	0.6395	0.7592	0.7353	0.7569	**0.7666**
	Avg.	0.5549	0.5538	0.5494	0.5361	0.5549	0.5522	0.5593	**0.5621**
RF	F194	0.4387	0.4401	0.4380	0.4437	0.4275	**0.4514**	0.4169	0.4479
	VOC	0.6253	0.6200	0.6129	0.6140	0.6259	0.6228	0.6307	**0.6321**
	SUN	0.6121	**0.6148**	0.6099	0.6096	0.6103	0.6144	0.6007	0.6135
	CLEF	0.6134	0.6273	0.6252	0.6294	0.6315	**0.6358**	0.5836	0.6304
	Cifar100	0.6055	0.6016	0.5976	0.6037	0.6034	0.6003	0.6047	**0.6060**
	ILSVRC	0.8380	0.8411	0.8142	0.8122	0.8485	0.8387	0.8470	**0.8517**
	Avg.	0.6212	0.6223	0.6159	0.6215	0.6250	0.6266	0.6139	**0.6281**

5.2. Comparison of local classification accuracy

We conduct experiments on imbalanced long-tailed datasets to verify the effectiveness of DAFS for tail classes. Table 7 reports the ACC values for the head, middle, tail, and all classes of the Cifar100-LT datasets. From the local classification accuracy results in Table 7, we can obtain the following conclusions:

(1) As the data imbalance rate increases, the overall and local class accuracy decreases. This indicates that imbalanced data distribution can lead to lower performance of tail classes.

(2) The performance of DAFS in the tail class is superior to that of other feature selection methods. These results demonstrate the effectiveness of DAFS in improving the accuracy of tail class classification.

5.3. Visualization analysis

We present a visualization of the data before and after feature selection using the t-SNE algorithm to demonstrate the effectiveness of the proposed method. After feature selection, the number of features reduces to 20% of the original. The visualizations before and after feature selection of the subtask for VOC are shown in Fig. 6. From this figure, the following conclusions can be obtained: (1) After dimensionality reduction using DAFS, the boundaries between classes become clearer, and the distances between classes increase. This indicates that removing redundant features enhances the separability of different classes. (2) Data points of the same class become closer, showing that the DAFS algorithm preserves the intra-class structure, leading to a denser distribution of data points within the same class in the feature space.

5.4. Ablation analysis

We discuss the influence of the weighted loss term and the head-tail feature equalization regularization term in the DAFS method and intuitively compare their effectiveness in improving the accuracy of tail classes using the following approach. (1) DAFS-BL combines the least squares loss function and the $l_{2,1}$-norm regularization term as baseline experience. (2) DAFS-CM adds weights based on sample distribution frequency to the loss function of DAFS-BL to minimize the risk of misclassification and improve the accuracy of tail class classification. (3) DAFS-HE adds a head-tail feature balancing regularizer based on DAFS-CM, which improves the classification performance of tail class samples by

Table 5
F_H comparison among seven methods on different classifiers (↑).

Classifier	Dataset	Method							
		HFSRR	HFSLDL	HFSCN	HFSDK	HFSLE	HFSCF	PHFS	DAFS
SVM	F194	**0.7164**	0.7153	0.7127	0.7052	0.7146	0.7164	0.6979	0.7157
	VOC	0.7975	0.7965	0.7969	0.7965	0.7959	0.7952	0.7968	**0.7998**
	SUN	0.8410	0.8423	0.8102	0.8363	0.8395	**0.8410**	0.8334	0.8384
	CLEF	0.7623	0.7643	0.7561	0.7513	0.7661	**0.7703**	0.7195	0.7689
	Cifar100	0.7877	0.7881	0.7859	0.7895	0.7895	0.7878	0.7854	**0.7934**
	ILSVRC	0.9587	0.9588	0.9516	0.9518	0.9605	0.9582	0.9588	**0.9605**
	Avg.	0.8106	0.8109	0.8022	0.8051	0.8110	0.8115	0.7986	**0.8128**
KNN	F194	0.6845	0.6836	0.6847	0.6803	0.6770	0.6843	0.6643	**0.6915**
	VOC	0.8148	0.8178	0.8155	0.8170	0.8178	0.8158	**0.8258**	0.8226
	SUN	0.8102	0.8099	0.8122	0.8057	0.8134	0.8104	0.8099	**0.8141**
	CLEF	**0.7787**	**0.7787**	0.7615	0.7733	0.7712	0.7779	0.7395	0.7701
	Cifar100	0.7748	0.7741	0.7717	0.7731	0.7754	0.7732	0.7788	**0.7843**
	ILSVRC	0.9285	0.9293	0.9084	0.9021	0.9339	0.9276	0.9330	**0.9360**
	Avg.	0.7986	0.7989	0.7923	0.7919	0.7981	0.7982	0.7919	**0.8031**
RF	F194	0.7613	0.7610	0.7556	0.7596	0.7540	0.7678	0.7408	**0.7688**
	VOC	0.8197	0.8190	0.8154	0.8161	0.8201	0.8183	**0.8238**	0.8235
	SUN	0.8389	0.8399	0.8374	0.8376	0.8386	0.8399	0.8343	**0.8403**
	CLEF	0.7630	0.7755	0.7713	0.7770	0.7744	**0.7801**	0.7475	0.7714
	Cifar100	0.7809	0.7786	0.7777	0.7792	0.7803	0.7783	0.7788	**0.7819**
	ILSVRC	0.9553	0.9559	0.9491	0.9487	0.9579	0.9552	0.9575	**0.9587**
	Avg.	0.8199	0.8217	0.8178	0.8197	0.8209	0.8233	0.8138	**0.8241**

Table 6
F_{LCA} comparison among seven methods on different classifiers (↑).

Classifier	Dataset	Method							
		HFSRR	HFSLDL	HFSCN	HFSDK	HFSLE	HFSCF	PHFS	DAFS
SVM	F194	0.6392	0.6379	0.6364	0.6284	0.6378	0.6390	0.6234	**0.6393**
	VOC	0.7628	0.7627	0.7617	0.7618	0.7616	0.7609	0.7636	**0.7660**
	SUN	0.7741	**0.7758**	0.7717	0.7692	0.7724	0.7743	0.7645	0.7710
	CLEF	0.7305	0.7329	0.7261	0.7180	0.7346	0.7401	0.6846	**0.7408**
	Cifar100	0.7667	0.7673	0.7645	0.7682	0.7685	0.7667	0.7642	**0.7724**
	ILSVRC	0.9226	0.9229	0.9093	0.9091	**0.9260**	0.9216	0.9235	0.9258
	Avg.	0.7660	0.7666	0.7616	0.7591	0.7668	0.7671	0.7539	**0.7692**
KNN	F194	0.6080	0.6052	0.6102	0.6038	0.5986	0.6069	0.5873	**0.6141**
	VOC	0.7805	0.7840	0.7807	0.7823	0.7842	0.7814	**0.7921**	0.7899
	SUN	0.7240	0.7235	0.7177	0.7175	**0.7278**	0.7243	0.7214	0.7274
	CLEF	0.7493	**0.7497**	0.7329	0.7432	0.7390	0.7467	0.7083	0.7371
	Cifar100	0.7519	0.7507	0.7475	0.7495	0.7528	0.7504	0.7568	**0.7617**
	ILSVRC	0.8656	0.8668	0.8259	0.8149	0.8759	0.8638	0.8744	**0.8798**
	Avg.	0.7465	0.7466	0.7358	0.7352	0.7464	0.7456	0.7401	**0.7516**
RF	F194	0.6935	0.6939	0.6905	0.6944	0.6861	**0.7011**	0.6761	0.7004
	VOC	0.7871	0.7849	0.7809	0.7815	0.7874	0.7855	0.7910	**0.7912**
	SUN	0.7673	0.7688	0.7657	0.7656	0.7665	**0.7687**	0.7606	0.7685
	CLEF	0.7357	0.7470	0.7434	0.7480	0.7479	**0.7526**	0.7173	0.7462
	Cifar100	0.7589	0.7565	0.7547	0.7575	0.7580	0.7559	0.7576	**0.7596**
	ILSVRC	0.9164	0.9178	0.9043	0.9033	0.9216	0.9166	0.9209	**0.9232**
	Avg.	0.7765	0.7782	0.7733	0.7750	0.7779	0.7801	0.7706	**0.7815**

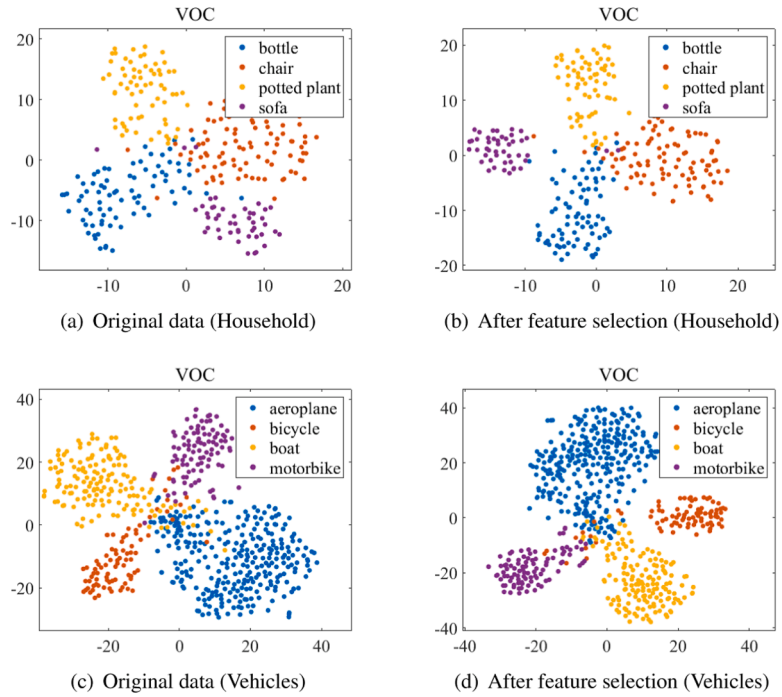
optimizing feature weights. Figs. 7 and 8 present the ACC values of the tail classes across three Cifar100-LT datasets using different classifiers.

From Figs. 7 and 8, we observe and summarize the following conclusions: (1) The tail class accuracy of DAFS-CM is higher than that of the baseline DAFS-BL. This indicates that the weighted loss effectively improves the accuracy of tail classes and minimizes the risk of misclassification through distribution-aware weight adjustments. (2) The performance of DAFS-HE on tail classes is superior to that of DAFS-CM. This demonstrates that the head-tail feature equalization regularization term further enhances the distinguishability of tail classes by constraining the feature distribution. (3) The tail class accuracy of DAFS-HE is significantly higher than that of DAFS-BL. This shows that combining the weighted loss term and feature equalization regularization term can jointly improve the accuracy of tail classes.

5.5. *Effectiveness analysis of feature differentiation regularizer*

We conduct experiments on the Cifar100-LT ($\rho = 50$) dataset to demonstrate the performance of feature differentiation regularization term in distinguishing easily confused classes. In the coarse-grained node “Human”, there are five easily confused classes: “Boy”, “Baby”, “Girl”, “Man”, and “Woman”. DAFS-ED adds the feature differentiation regularization term based on DAFS-BL to improve the accuracy of the easily confused classes. The experimental results of DAFS-ED in these classes are shown in Table 8.

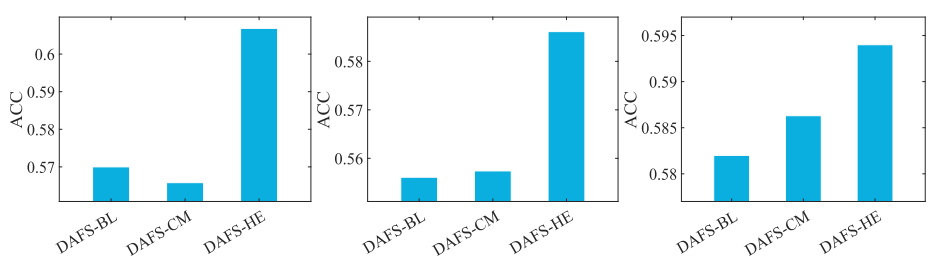
From Table 8, we can obtain: (1) DAFS-BL has a lower ACC in the “Woman” class, indicating that its samples are not sufficiently distinct from other easy-confused classes, resulting in lower accuracy. (2) The ACC value of the “Woman” class significantly increased from



(a) Original data (Household) (b) After feature selection (Household)

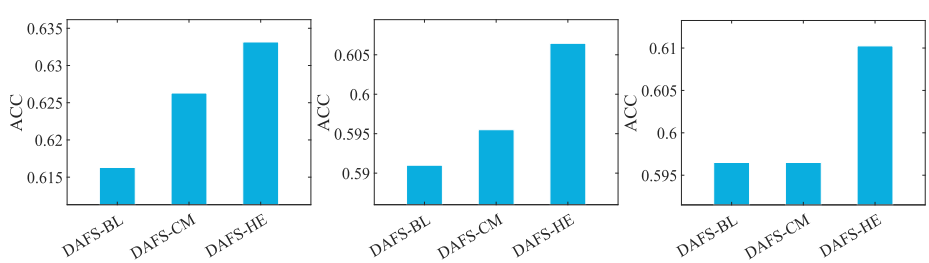
(c) Original data (Vehicles) (d) After feature selection (Vehicles)

Fig. 6. t-SNE visualization of the VOC dataset before and after feature selection.



(a) Cifar100-LT (ρ = 50) (b) Cifar100-LT (ρ = 100) (c) Cifar100-LT (ρ = 200)

Fig. 7. Ablation analysis of Cifar100-LT datasets on the KNN classifier.



(a) Cifar100-LT (ρ = 50) (b) Cifar100-LT (ρ = 100) (c) Cifar100-LT (ρ = 200)

Fig. 8. Ablation analysis of Cifar100-LT datasets on the RF classifier. .

24% in DAFS-BL to 43% in DAFS-ED, increasing 19 percentage points. This result demonstrates the effectiveness of the feature differentiation regularization term in improving the classification accuracy of easily confused classes.

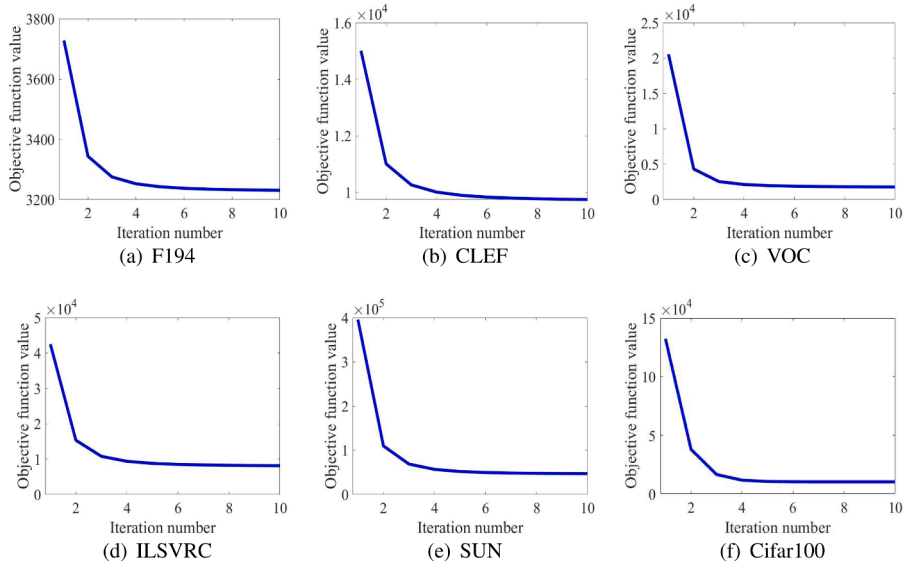


Fig. 9. Convergence curves for the DAFS.

Table 7
Comparison of local classification accuracy on Cifar100-LT with different ρ.

Method	$\rho = 100$			
	Tail (31)	Middle (34)	Head (35)	All (100)
HFSRR	0.5623	0.6079	0.5980	0.5903
HFSLDL	0.5706	0.6085	0.5980	0.5931
HFSCN	0.5597	0.5994	0.6023	0.5881
HFSDK	0.5519	0.5994	0.5891	0.5811
HFSLE	0.5919	0.6209	0.5991	0.6043
HFSCF	0.5690	0.6059	0.5957	0.5909
PHFS	0.5681	0.5947	0.5574	0.5734
DAFS	**0.5948**	**0.6250**	**0.6134**	**0.6116**
Method	$\rho = 200$			
	Tail (40)	Middle (30)	Head (30)	All (100)
HFSRR	0.5717	0.6123	0.5880	0.5904
HFSLDL	0.5737	0.6107	0.5913	0.5918
HFSCN	0.5603	0.6173	0.5865	0.5879
HFSDK	0.5707	0.6027	0.5843	0.5857
HFSLE	0.5723	0.6183	0.5930	0.5944
HFSCF	0.5730	0.6110	0.5860	0.5896
PHFS	0.5583	0.5907	0.5558	0.5670
DAFS	**0.5833**	**0.6227**	**0.6018**	**0.6025**

Table 8
Classification accuracy (%) of KNN classifier on five easily confused classes.

Method	Boy	Baby	Girl	Man	Woman
DAFS-BL	48	31	39	53	24
DAFS-ED	**54**	**32**	**45**	**56**	**43**

5.6. Convergence analysis

In this section, we analyze the convergence of the DAFS method. The maximum number of iterations T is set to 10 for the experiments. Fig. 9 shows the convergence curves for all datasets based on the value of the objective function in Eq. (7). This figure shows that the objective function value monotonically decreases and converges to a stable value within ten iterations for all datasets.

6. Conclusions and future work

We proposed a distribution-aware hierarchical feature selection method for long-tailed classification (DAFS). We integrated sample distribution information and inter-class similarity into the loss function and regularization terms. This approach effectively addresses the challenges of class imbalance and distinguishing easily confused classes in long-tailed data. Specifically, we introduced a distribution-aware penalty coefficient to weight the loss and incorporate it into the regularization term, thereby reducing the misclassification risk of tail classes and improving the classification accuracy of tail classes. Additionally, we designed a feature differentiation regularization term to enhance the distinguishability of easily confused classes, which leverages multi-granularity knowledge and class similarity. We demonstrated the effectiveness of DAFS and its superiority over other feature selection methods through extensive experiments on long-tailed benchmark datasets.

In this work, our method addresses the long-tailed problem by adjusting weights without altering the original sample distribution. However, there are still some limitations: (1) DAFS assumes that all labels are accurate and no outliers are in the data. Unfortunately, real-world datasets often contain incorrectly labeled instances or outliers, which can introduce noise and pose challenges to classification tasks. (2) DAFS does not consider the risk of cross-hierarchical misclassification in tree-structured data, which may negatively impact the performance of datasets with complex tree structures. In future work, we will explore the progressive learning strategy to mitigate the effects of incorrect labels, and design penalty weights based on hierarchical structures to address cross-layer misclassification issues.

CRediT authorship contribution statement

Yang Zhang: Writing – review & editing, Writing – original draft, Methodology; **Jie Shi:** Supervision, Writing – review & editing; **Yanfang Liu:** Supervision, Writing – review & editing; **Hong Zhao:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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